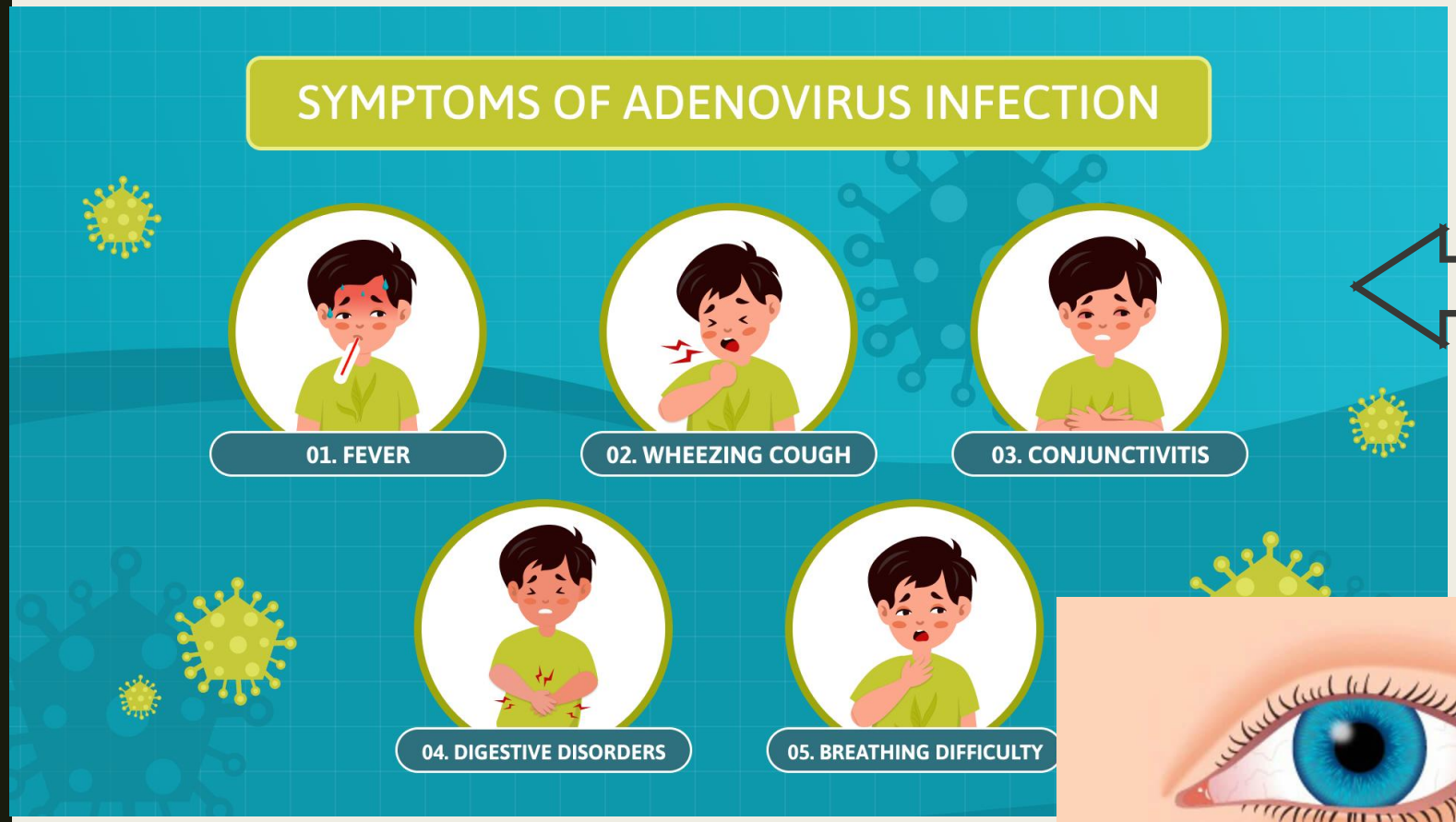
ACUTE BRONCHIOLITS

- Lower resp tract viral infection affecting <2yr olds.
- Peak incidence between 2-6 month of age.
- Characterized by URTI preceeding it.
- Typically presents with rhinorrhea, fever, cough, wheezes +/- creps, respiratory distress, +/- conjunctivitis.

- **Risk Factors:**
  - Prematurity <36 weeks
  - <3 months old
  - Airway anatomical abnormality
  - Immunodeficiency
  - Low birth weight
  - CLD
  - Congenital heart disease
  - Neurological disease

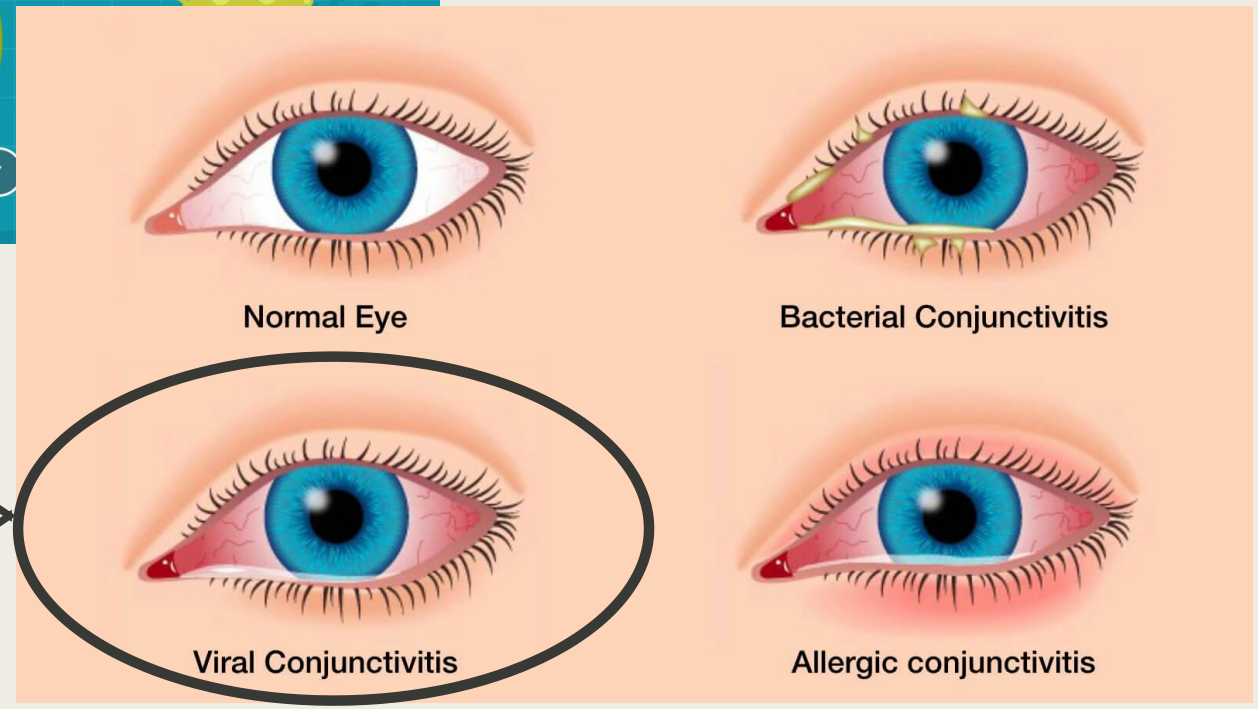
Viral causes in order of incidence

- **RSV** – peak in fall and winter.
- **Rhinovirus** – peak in spring and fall.
- **Parainfluenza virus** – type 3 peaks in spring and fall. Type 1&2 more common with croup.
- **Human metapneumovirus** – peaks in March and April, usually occurs with other viruses.
- **Influenza virus** – A and B subtype are treatable with Tamiflu
- **Adenovirus** – may also cause pharyngitis, more perulent secretions
- **Coronaviruses** – SARS-CoV-2, SARS-Cov-1, MERS-Cov.
- **Human bocavirus** – peaks in fall and winter, can cause pertussis-like illness.



Plenty of secretions!

Conjunctivitis with watery discharge. May be crusting.



Flu A & B have markedly elevated temperatures that partially respond to antipyretics. Generalized body aches, malaise, headache.

Clinical Features

- Fever, cough, resp distress following 1-3 days of URTI
- Self-limiting, most of unhospitalized children recover by 28 days.
- Peak of illness is day 3-5 then gradually resolve over weeks.
- Average length of hospitalization 2-5 days.
  Shortest in Rhinovirus, longest in RSV.
- Course is longer in those of high risk group.
- Wheezes resolve in 1 week. Cough takes 2-3 weeks to resolve.

COMPLICATIONS

- Mostly occurs to high risk group (apnea + resp failure)

- **Dehydration:** increased insensible losses, decreased intake, +/- vomitting.
- **Aspiration pneumonia**: cough and tachypnea with full stomach.
- **Apnea**: mostly in high risk group + RR <30 or >70, age <2months, on O_2.
  Risk factor to develop Resp Failure. No virology corelation.
- **Resp Failure**: More in high risk group and age <6weeks.
  Mucus plugs & atelectasis are common finding, resolve fast.
  Hypercapnic Resp Failure with fatigue require resp support.
- **Secondary bacterial infection**: uncommon 1.2%. Higher in those in PICU.
  Usually seen as abrupt worsening in fever and severity after several days of static course.

Bronchiolitis Obliterans

- Post-infectious BO usually (not exclusively) due to adenovirus infection.
- Results in diffuse lung disease (ILD)
- High resolution CT shows mosaic perfusion, air trapping, and bronchiectasis
- Predominantly affect gas exchange in the lung tissue
- Most need oxygen

DIAGNOSIS

- Made clinically.
- CXR and labs are only for severe, complicated, suspiciouns of another daignosis (hear failure, pneumonia, foreign body, etc).
- Nasopharyngeal swab to identify treatable causes Flu A + Flu B.
- Blood gas analysis when you suspect respiratory failure.

Management

- Mainly supportive: oxygenation + hydration
- **OUTPATIENT**: Adequate hydration
  Normal saline drops + suction
  Antipyretic
  +/- tamiflu
- **INPATIENT**: Hydration with IVF
  Normal saline drops + suction
  +/- tamiflu
  Bronchodilators if responds to trial
  3% saline neb → poor evidence, not routinely advised, may increase cough. Waranted trial in severe cases
  Glucocorticoids only in recurrent wheezy chest or family hx of asthma
  Respiratory support → HFNC, CPAP, M.V